PROJECT REPORT

SAP MM Implementation for Procure-to-Pay (P2P) Process in KIIT University

Title: End-to-End Procure-to-Pay (P2P) Process in SAP MM

Problem Statement:

An organization needs an organized procurement process that involves handling all the procurement cycle starting from identification of the materials required till making the payment to the supplier. Some of the challenges associated with traditional manual procurement include:

• Manual purchasing requisition resulting into delayed processes and unauthorized purchases

• Difficulty in assessing the supplier performance and comparing offers by suppliers

• Quantity mismatch between order, receipt and invoicing

• Poor inventory control and stock-outs

• Lack of effective data flow for payment process in the finance department

Such problems result in increased operating expenses and difficulties in managing supplier relationships hence the requirement of ERP system that can automate procurement operations effectively.

Solution Overview:

The SAP Materials Management (MM) Procure-to-Pay (P2P) helps organizations automate the entire procurement process.

Main Steps in Procurement Cycle Using P2P:
• Requirement identification
• Vendor selection/source determination
• Purchase order processing
• Goods Receipt• Invoice verification
• Transaction Processing

What is more important, the SAP Material Management is fully integrated with other modules including:
• SAP Financial Accounting (FI) - General Ledger and Vendor Payments.
• SAP Sales & Distribution (SD) for sales order created from procured materials.
This integration allows for a seamless data flow between the supply chain and financial systems.

Step-by-Step Process:

Step 1: Requirement of Requirements (Transaction-Code: ME51N)

- A requirement arises for particular materials/services in the department.
- The internal Purchase Requisition (PR) order is created for the material to be Procured.

Step 2: Source Selection and Vendor Evaluation (Transaction-Codes: ME41, ME47, ME49)

- The purchasing department makes the sources for materials.
- Request for Quotations (RFQs) are sent to multiple vendors.
- The quotations from various vendors are evaluated in the system, taking the Price into account.

Step 3: Creation of Purchase Order (Transaction-Code: ME21N)

- A Purchase Order (PO) is made based on RFQ or PR.
- The PO is a contract created with the vendor containing all the details of the material and its costs.

Step 4: Goods Receipt (Transaction-Code: MIGO)

- The purchased material is received from the vendor and Goods Receipt (GR) is done.
- The stock is updated, and both a material and an accounting document is generated in the system..

Step 5: Matching of Invoice and Documents (Transaction-Code: MIRO)

- The vendor sends invoices regarding the GR.
- PO, GR, and invoice are matched in the system to verify the quantity of goods delivered and cost.

Step 6: Invoice Payment (Transaction-Code: F-53)

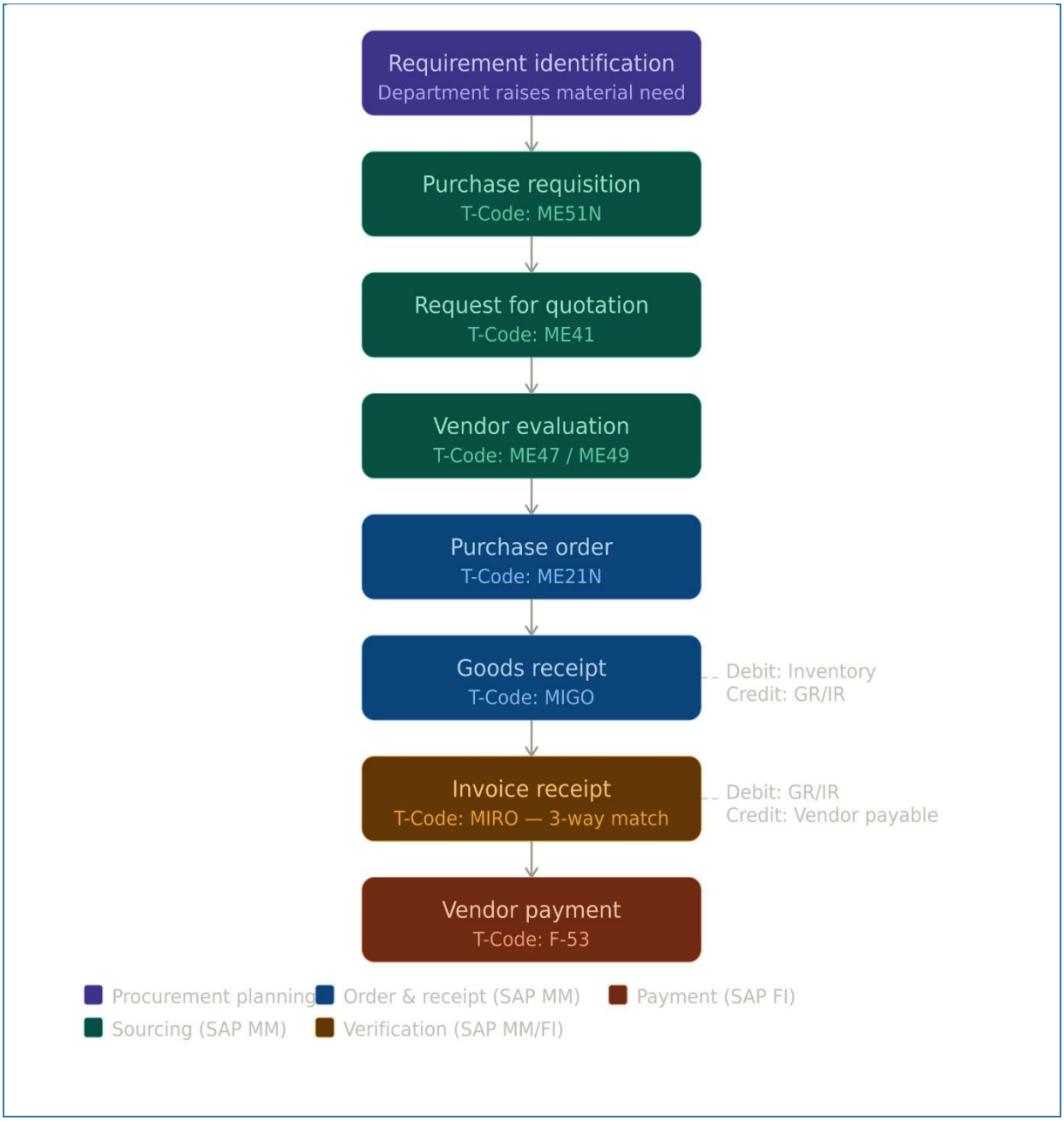
- The invoice sent by the vendor is validated by the finance department.
- The payment is made to the vendor.

Tech Stack:

- ERP System: SAP S/4HANA
- Module Used: SAP MM (Materials Management)
- Supporting Module: SAP FI (Financial Accounting)
- Database: HANA Database

- Interface: SAP GUI / FIORI Apps

PROCURE-TO-PAY FLOWCHART



Organizational Structure:

Component	Value
Client	900
Company Code	1000
Purchasing Organization	1000
Purchasing Group	001
Plant	1001
Storage Location	0001

Integration with Finance (FI):

The P2P process heavily integrates with SAP FI for financial postings at two major milestones:

Step	Debit Entry	Credit Entry
Goods Receipt (MIGO)	Debit: Inventory Account	Credit: GR/IR Clearing Account
Invoice Verification (MIRO)	Debit: GR/IR Clearing Account	Credit: Vendor Payable Account

Example Scenario:

For example, the KIIT IT Department needs to buy 50 new laptops for a lab upgrade:

• The IT lead generates a Purchase Requisition for 50 laptops.

• The purchasing team issues RFQs to Dell, HP and Lenovo.

• Purchase Order is raised to Lenovo as Lenovo gives best price.

• The 50 laptops are supplied by Lenovo and the warehouse team will record a Goods Receipt to add the stock.

• Lenovo raises an invoice of ₹25,00,000. Finance posts this invoice against the PO using MIRO.

• Finance operates the payment program to move the funds into Lenovo's bank account.

This demonstrates the whole, controlled cycle of buyin

Striking Points of the Project:

- Complete automation of the procurement lifecycle
- Data being managed centrally
- Improved purchase order accuracy and operational efficiency
- Real-time tracking of inventory and procurement data
- Seamless integration with the finance module
- Reduction in manual errors and maverick spending

Scopes for Improvements:

• Automated Purchase Order creation using SAP Fiori Apps and RPA (Robotic Process Automation).

• Deploy predictive analytics to anticipate material shortages before they occur.

• Shorter RFQ response times through vendor self-service portals.

CONCLUSION

The Procure-to-Pay (P2P) process in SAP MM is a very organised and automated way for a business to buy things. The system stops money from leaking out and makes sure that the data is very accurate by setting up a strict three-way matching system (PO, Goods Receipt, Invoice). Combining procurement and financial systems makes sure that everything is clear, follows the rules, and works better.

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